



POLITECNICO
MILANO 1863

LSM6DSOX Linux Driver Development

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Project overview

What we started from:

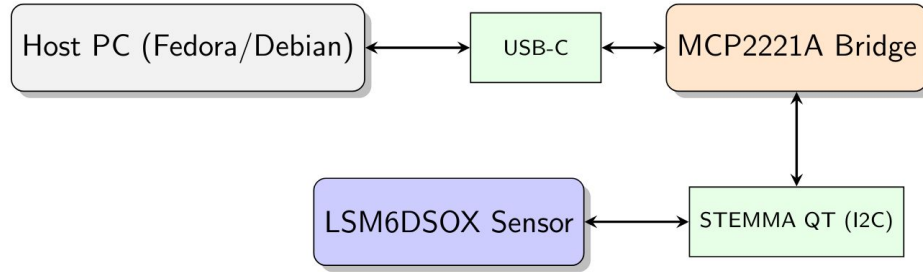
- Sensor: ST LSM6DSOX (6-axis IMU + Temperature).
- Hardware Setup: Adafruit breakout + MCP2221A (I2C-USB bridge).
- OS Environment: Virtualized Debian 13 (hrtimer support).
- Standard driver in the linux kernel

What we wanted to do:

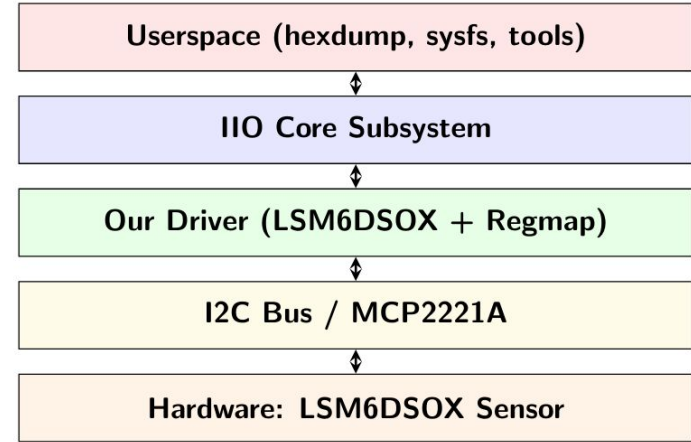
Make our own driver to read sensor data with the following functionalities:

- IIO Direct Mode (Sysfs).
- Triggered Buffer (kfifo) for high-speed streaming.

Hardware setup and IIO Subsystem Interaction

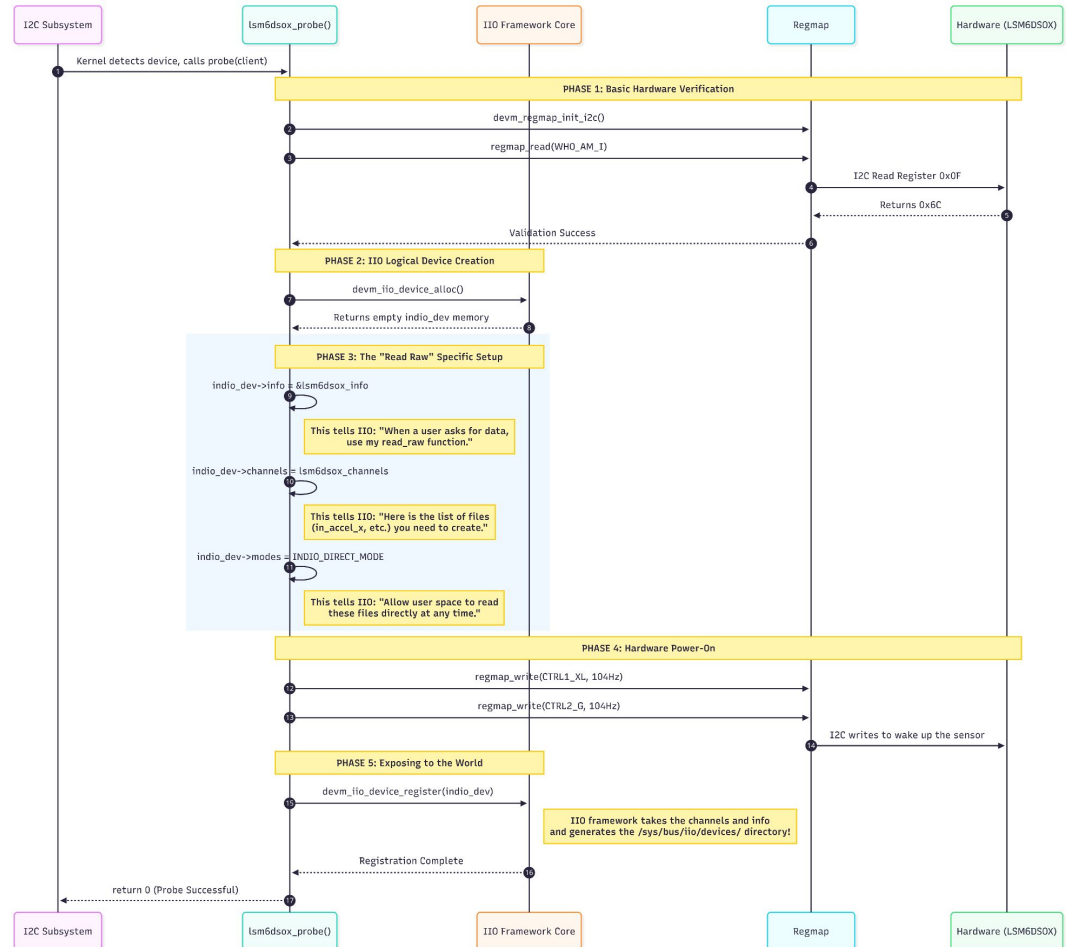


- **Connectivity:** Bridge provides I2C interface over USB.
- **Addressing:** SDO pulled to GND → Address 0x6a.
- **VM Passthrough:** USB device attached directly to the QEMU/KVM guest.

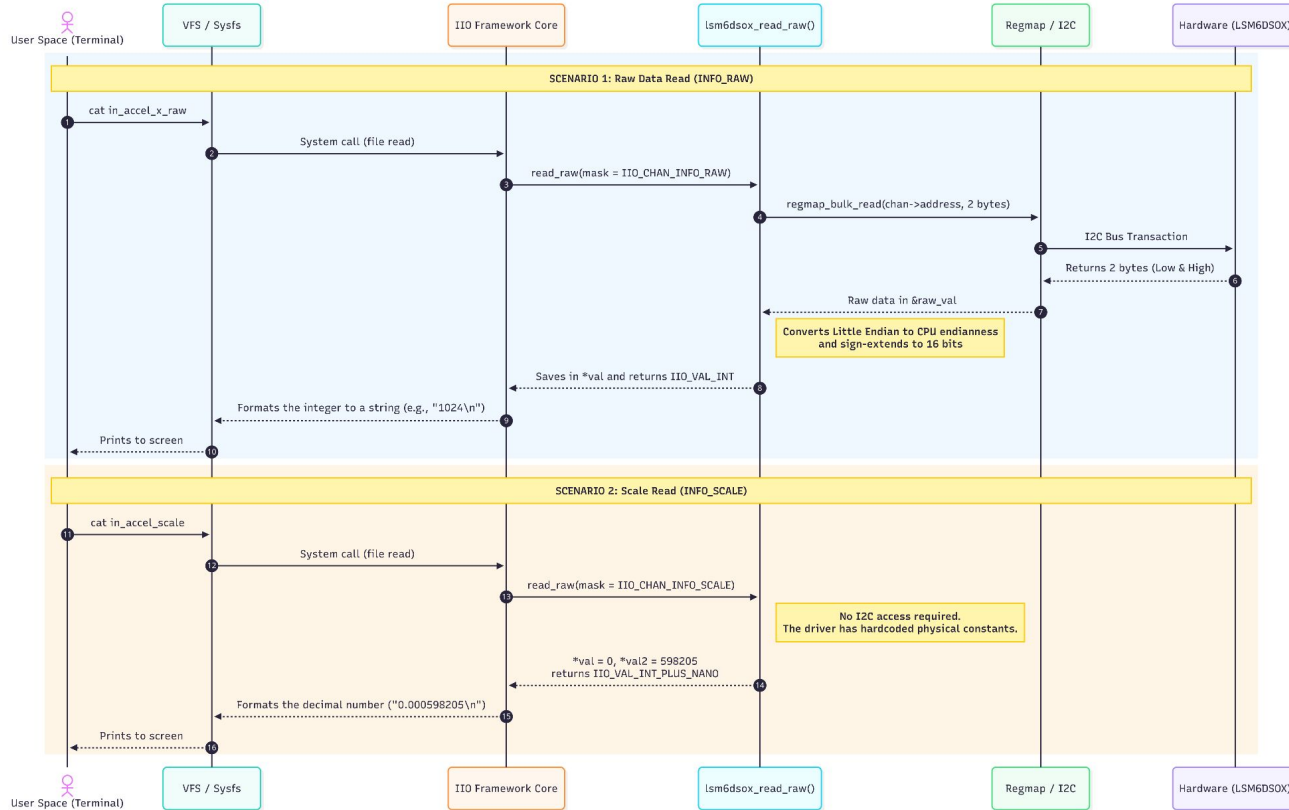


Direct Mode: read raw handles on-demand sysfs reads.

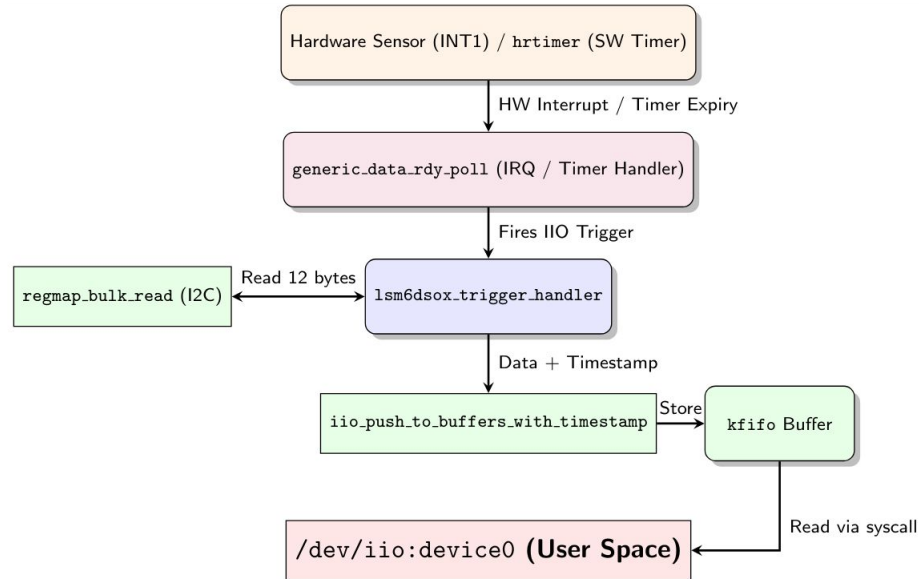
Probe function



Raw data read



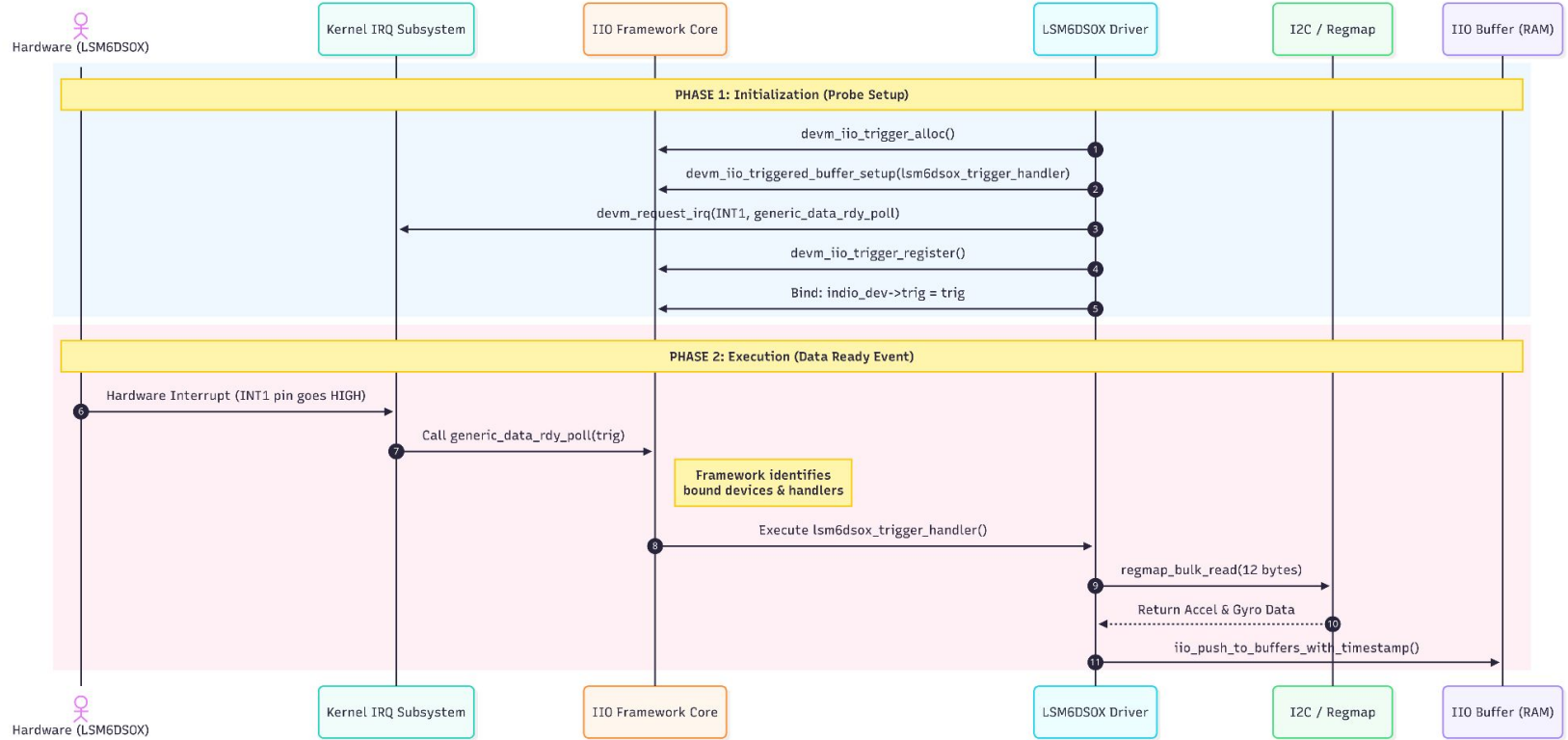
Triggered Buffer Data Flow



Atomic Acquisition: Reads all 6 axes in one burst.

Pushing: Data is timestamped and queued into the IIO FIFO.

Sequence Diagram



Comparison with Official Driver

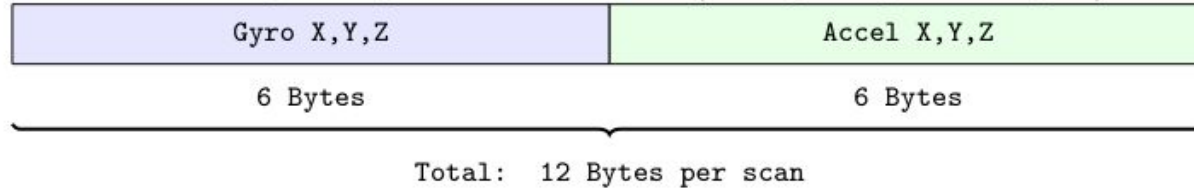
Feature	Our Driver	Official st_lsm6dsx
Transport	I2C only	I2C + SPI
Multi-device support	LSM6DSOX only	15+ variants
ODR / FS configuration	Hardcoded at probe	Runtime via sysfs
Data acquisition	Triggered buffer (kfifo)	Hardware FIFO + watermark
Scan mask optimization	Full burst always	Adapts to active channels
Device enumeration	Manual (new_device)	Device Tree / ACPI
Regmap	✓	✓
Advanced (MLC/FSM)	×	✓

Live Demo

Operational Demonstration

```
sudo cat /dev/iio:device0 | hexdump -C
```

Hexdump Output Breakdown (12 bytes per sample):



Outcomes

- Successful implementation of a complete IIO driver.
- Transition from Polling to Reactive (Interrupt-driven) design
- Successful implementation of *regmap* abstraction
- Code quality and kernel compliance with *checkpatch.pl*

Problems encountered:

- IIO Trigger Availability in VM
- Virtiofs Permissions
- Debugging and Logs

Learning outcomes: `

- Hardware-Software Interfaces in the Linux Kernel
- Hardware Abstraction using the *regmap* API
- Asynchronous Events and Triggers in the Industrial I/O (IIO) Framework
- Linux Kernel Modules and Drivers Development Workflow
- Sensor Data Processing and Unit Conversion